

Nicolás Francinelli

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Professional experience

Airbus OMR by ALTEN

Toulouse, France

Consultant – Robotics Engineer – Permanent Contract

03/2026 – Present

Development within the Primer Structural Defects project. Defect recognition on plane fuselage.

- Programming on Siemens Industrial PC. TIA Portal configuration and Profinet connection.
- Implementation of a process control system based on a Behavior Tree with connection to an HMI via a TCP/IP server and to a PLC using shared memory.

Magellium by LR Technologies

Toulouse, France

Consultant – Robotics Engineer – Permanent Contract

09/2025 – 12/2025

Development within the MMX space project for CNES.

- Development of an automated validation framework in C/Python for multi-platform test campaigns.
- Automation of code coverage analysis, memory analysis, log parsing, and results visualization.
- Development of 70+ unit tests and contribution to improving software quality and documentation.

Illumo Robotics

Toulouse, France

Robotics Engineer – Permanent Contract

12/2023 – 05/2025

Startup developing industrial pick-and-place solutions based on object recognition and AI.

- Development of Gazebo plugins to improve the simulation of a Yaskawa robotic arm and its actuators.
- Migration of existing code from ROS to ROS2. Design and implementation of ROS 2 nodes for robotic control, trajectory planning, and behavior management using state machines and behavior trees.
- Integration of industrial hardware (FRAMOS Ethernet camera, IFM IO-Link master).
- Delivery of 5+ robotic safety training sessions to six-person teams in Spain.

Capgemini Engineering

Toulouse, France

R&D Robotics Engineer – Internship

02/2023 – 08/2023

Simulation of an autonomous mobile robot for electric vehicle charging.

- Implementation of an autonomous navigation system for dynamic obstacle avoidance.
- Object position estimation using a camera. Admittance-based control of a UR 10e robotic arm.

Education

École Centrale de Nantes

Nantes, France

Master in Advanced Robotics CORO IMARO

09/2021 – 08/2023

CGPA 3.49/4. Student representative of Master Control and Robotics, 2 years.

Universidad Nacional de La Plata

La Plata, Argentina

Degree in Electronics Engineering

03/2016 – 02/2021

Average 8.77/10 – Awarded “Best Graduate 2020”. Teaching Assistant at Circuit Theory I, 2 and a half years.

Hard Skills

C, C++, Python, ROS/ROS2, Gazebo, Git, Docker, OpenCV, Linux, CUnit, Machine Learning, Behavior Trees, CAD

Projects

Universidad Nacional de La Plata – LEICI

La Plata, Argentina

Final Degree Project

01/2020 – 02/2021

- Implementation of an autonomous navigation system using a Convolutional Neural Network (CNN).
- Implemented a Kalman filter for data fusion between the odometry and the IMU.
- Implementation of a PID controller to correct robot deviation errors during direction tracking.

Soft Skills & Interests

Languages: English (C2), French (C2), Spanish (Native), Italian (A2).

Certifications: CodingGame Certification – C++ (2022).

Interests: Cooking, rugby, F1.